



PSYCH 201B

Statistical Intuitions for Social Scientists

Statistical Inference (Discussion)

Today's Plan

1. Group Discussion on Readings
2. Break
3. Statistical Power & Parameter Inference

Major Themes (I think are important)

- Models as error-reducing devices
- Estimators as random variables
- Assumptions should guide decisions
- The pathology of Null-Hypothesis-Significance-Testing (NHST)

What is statistical power?

...some background first

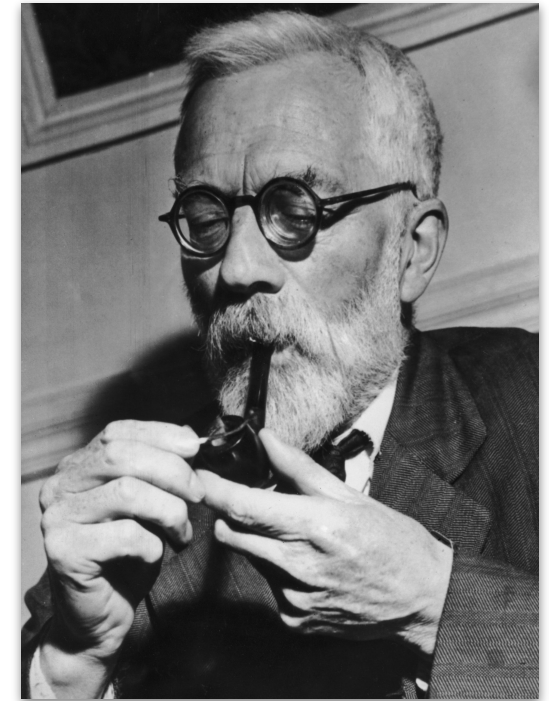
Perspective 1: P-values are **evidence** for hypotheses

*"If P is between .1 and .9 there is certainly no reason to suspect the hypothesis tested. If it is below .02 it is strongly indicated that the hypothesis fails to account for the whole of the facts. **We shall not often be astray if we draw a conventional line at .05** ... it is convenient to draw the line at about the level at which we can say: Either there is something in the treatment, or a coincidence has occurred such as does not occur more than once in twenty trials..."*

BUT

*"**no scientific worker has a fixed level of significance** at which from year to year, and in all circumstances, he rejects hypotheses; he rather gives his mind to each particular case in the light of his evidence and his ideas"*

"invented" $p < .05$



Sir Ronald **F**isher

No! P-values are **not** evidence for hypotheses

*"no test based upon a theory of probability can by itself provide any valuable evidence of the truth or falsehood of a hypothesis. But we may look at the purpose of tests from another viewpoint. **Without hoping to know** whether each separate **hypothesis is true or false**, we may search for **rules** to govern our behavior with regard to them, in following which we insure that, **in the long run of experience, we shall not often be wrong**"*

Dad (Karl Pearson) invented... like everything
r correlation coefficient, Chi-square, PCA ...

invented
confidence intervals



Jerzy Neyman Egon Pearson

We can't know which specific decisions are right or wrong...
but we can **control how often** our decisions will be **wrong in the long run**

*"no test based upon a theory of probability can by itself provide any valuable evidence of the truth or falsehood of a hypothesis. But we may look at the purpose of tests from another viewpoint. **Without hoping to know** whether each separate **hypothesis is true or false**, we may search for **rules** to govern our behavior with regard to them, in following which we insure that, **in the long run of experience, we shall not often be wrong**"*

Dad (Karl Pearson) invented... like everything
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Jerzy Neyman Karl Pearson

Neyman-Pearson Approach

- Two possible states of the world
 - H_0 (null) is **true**
 - H_0 (null) is **false**
- Two decisions we can make
 - **reject** H_0
 - **retain** H_0
- Two ways we can be wrong
 - **Reject** when H_0 is **true**
 - **Retain** when H_0 is **false**

False Positive **Type I** error α

False Negative **Type II** error β



Imagine you're a (medical) doctor...

H_0 : Not pregnant H_1 : Pregnant

Type I Error



Type II Error



Type I Error: Falsely rejecting the null hypothesis (even though it is true)

Type II Error: Failing to reject the null hypothesis (even though it is false)

Imagine you're a (medical) doctor...

H_0 : Not pregnant H_1 : Pregnant

Ways to
be wrong



false
positive **Type I error** α
 $p(\text{reject } H_0 | H_0 \text{ is true})$

false
negative **Type II error** β
 $p(\text{not reject } H_0 | H_1 \text{ is true})$

Ways to
be right



true
positive **Sensitivity** **Power**
 $p(\text{reject } H_0 | H_1 \text{ is true})$ $1 - \beta$

true
negative **Specificity** $1 - \alpha$
 $p(\text{not reject } H_0 | H_0 \text{ is true})$

The **power** of a binary hypothesis test is the probability that the test rejects the null hypothesis (H_0) when a **specific** alternative hypothesis (H_1) is true.

H_0 : Students and non-students have the same balance.

Model C

$$Y_i = \beta_0 + \epsilon_i$$

$$\beta_1 = 0$$

H_1 : Students and non-students have different balances.

Model A

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

$$\beta_1 \neq 0$$

We cannot calculate power in this case.
We need a **specific** alternative hypothesis!

The **power** of a binary hypothesis test is the probability that the test rejects the null hypothesis (H_0) when a **specific** alternative hypothesis (H_1) is true.

H_0 : Students and non-students have the same balance.

Model C

$$Y_i = \beta_0 + \epsilon_i$$

$$\beta_1 = 0$$

H_1 : Students and non-students have different balances.

Model A

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

$$\beta_1 = 300$$

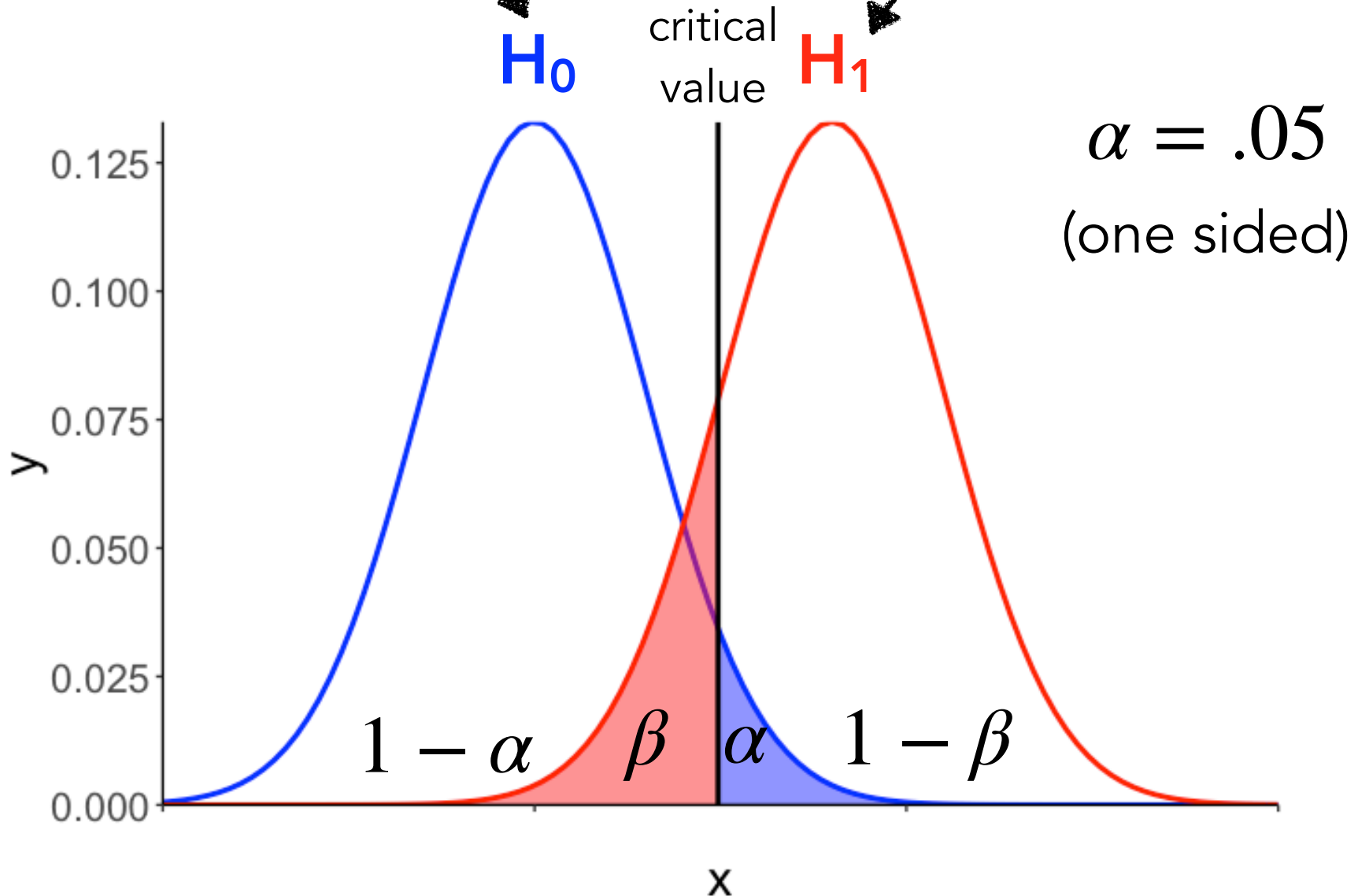
We can calculate power in this case since we have a **specific** alternative hypothesis!

What affects power?

false positive	Type I error $p(\text{reject } H_0 H_0 \text{ is true})$	α	true positive	Sensitivity $p(\text{reject } H_0 H_1 \text{ is true})$	Power $1 - \beta$
false negative	Type II error $p(\text{not reject } H_0 H_1 \text{ is true})$	β	true negative	Specificity $p(\text{not reject } H_0 H_0 \text{ is true})$	$1 - \alpha$

sampling distribution if H_0 is true

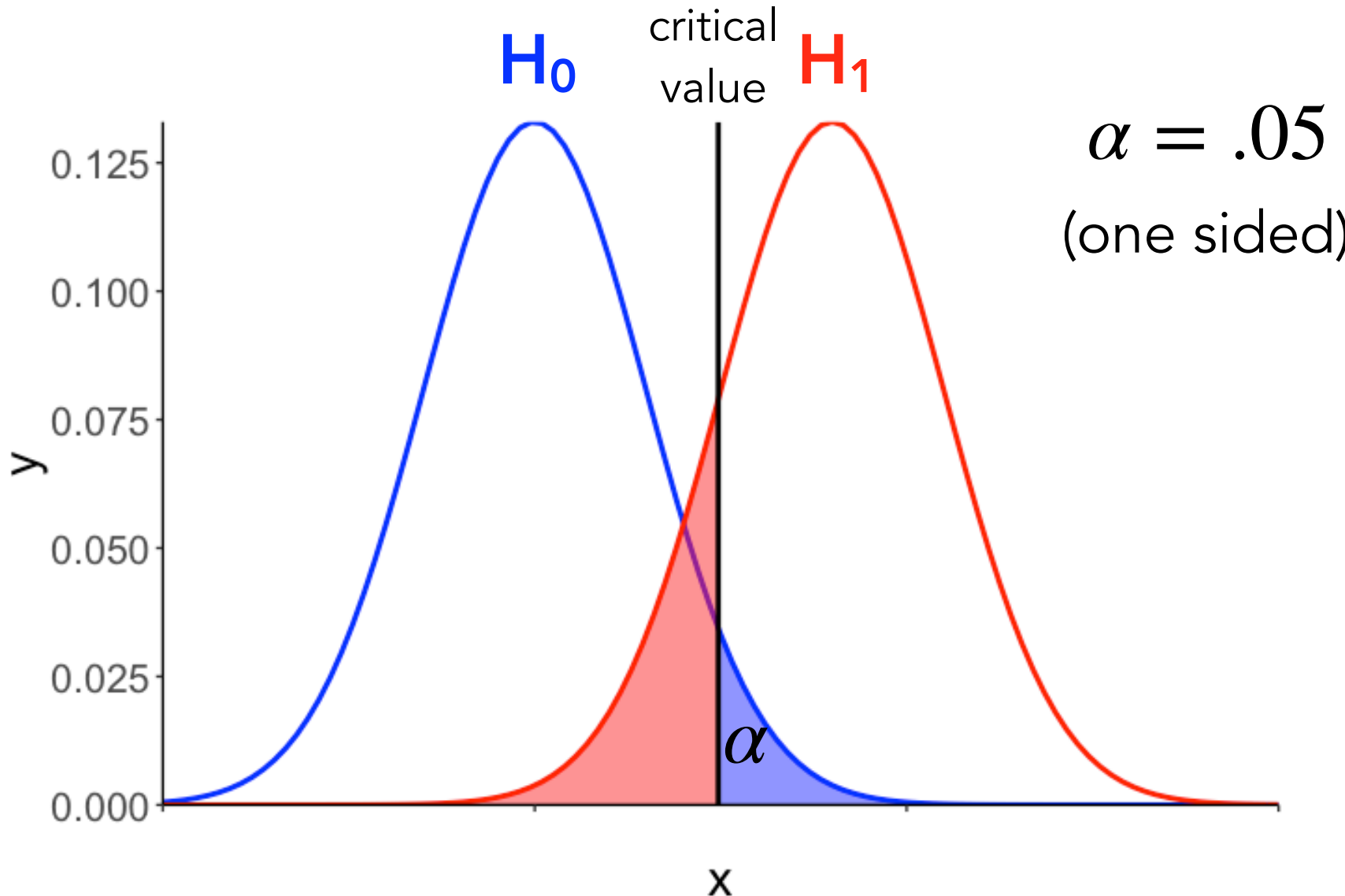
sampling distribution if H_1 is true



α affects power

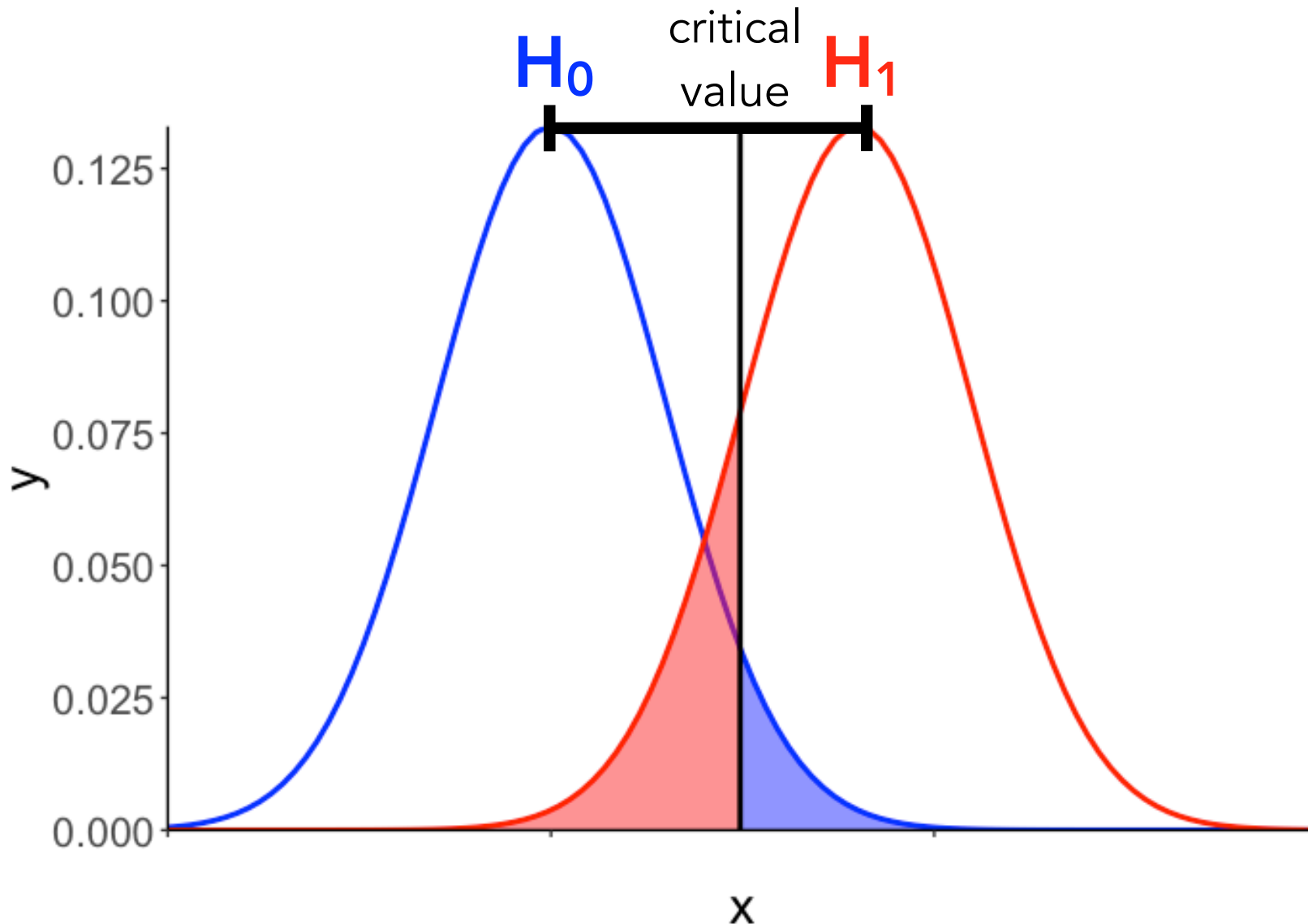
(e.g. p-value threshold)

false positive	Type I error $p(\text{reject } H_0 H_0 \text{ is true})$	α	true positive	Sensitivity $p(\text{reject } H_0 H_1 \text{ is true})$	Power $1 - \beta$
false negative	Type II error $p(\text{not reject } H_0 H_1 \text{ is true})$	β	true negative	Specificity $p(\text{not reject } H_0 H_0 \text{ is true})$	$1 - \alpha$



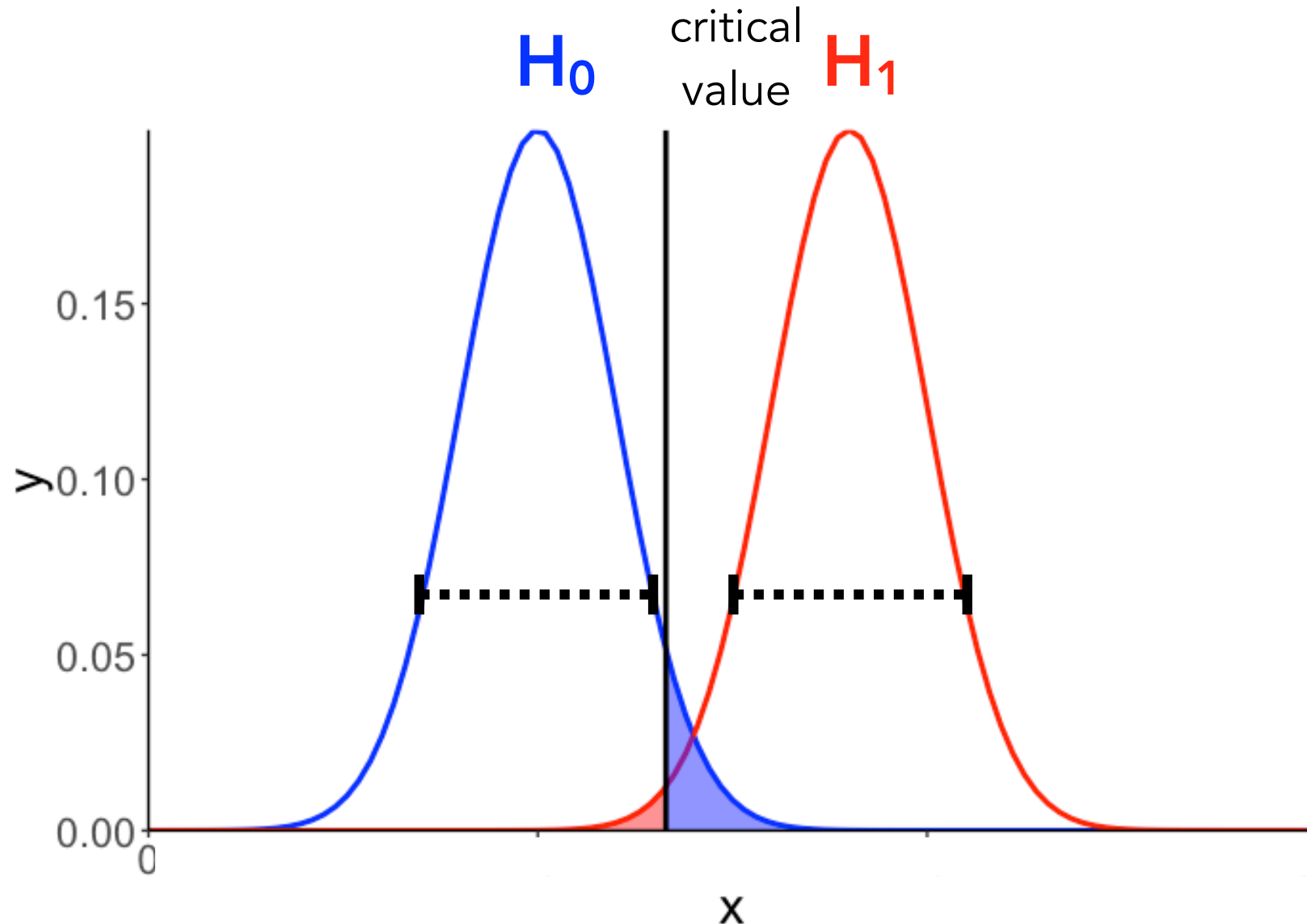
effect size affects power

false positive	Type I error $p(\text{reject } H_0 H_0 \text{ is true})$	α	true positive	Sensitivity $p(\text{reject } H_0 H_1 \text{ is true})$	Power $1 - \beta$
false negative	Type II error $p(\text{not reject } H_0 H_1 \text{ is true})$	β	true negative	Specificity $p(\text{not reject } H_0 H_0 \text{ is true})$	$1 - \alpha$

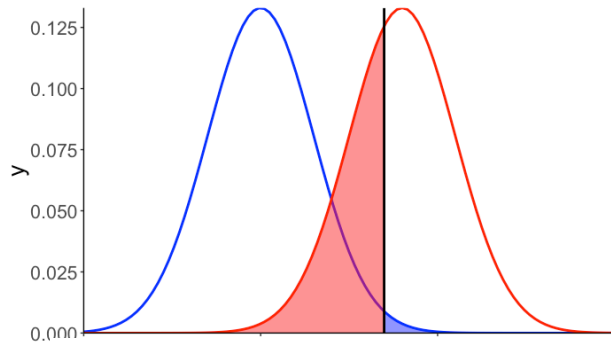
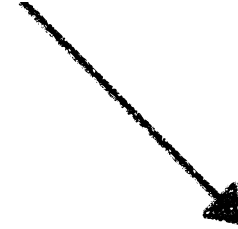
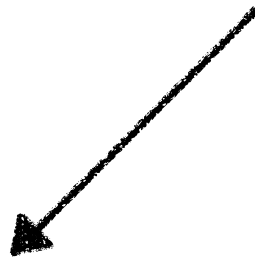
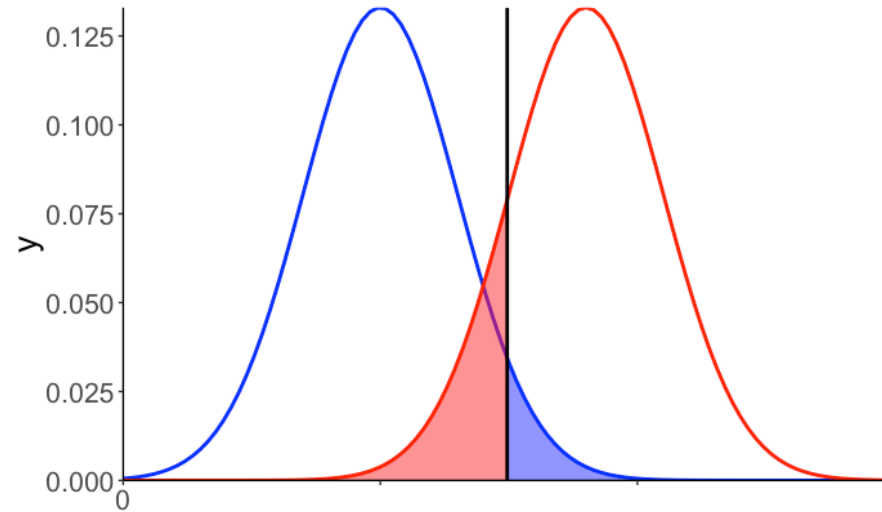


variance affects power

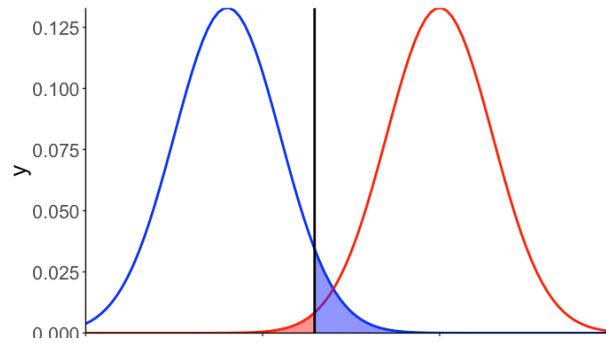
false positive	Type I error $p(\text{reject } H_0 H_0 \text{ is true})$	α	true positive	Sensitivity $p(\text{reject } H_0 H_1 \text{ is true})$	Power $1 - \beta$
false negative	Type II error $p(\text{not reject } H_0 H_1 \text{ is true})$	β	true negative	Specificity $p(\text{not reject } H_0 H_0 \text{ is true})$	$1 - \alpha$



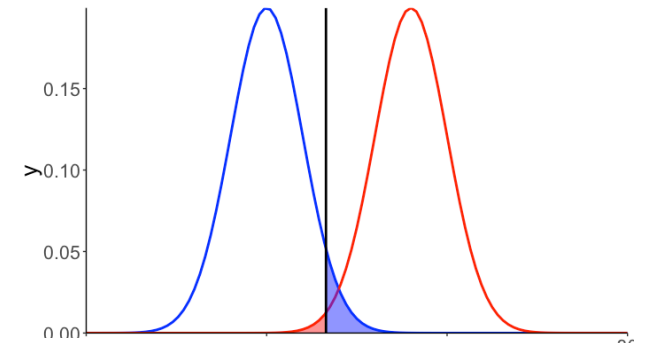
The knobs we can turn to affect power



a threshold



effect size



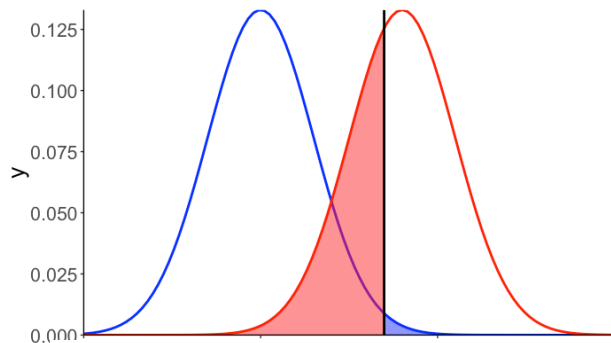
sample size

The knobs we can turn to affect power

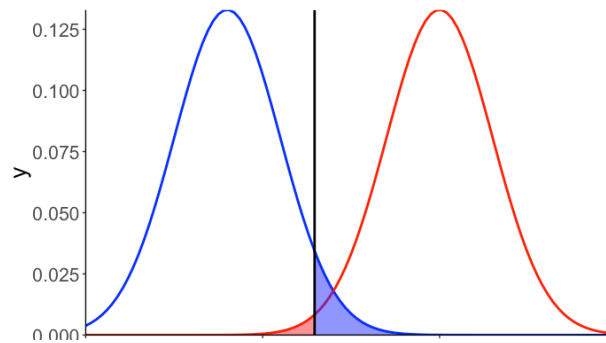
...how much **data** do I need to collect to have a good chance of **detecting** an effect at least X large?

...how much does my estimation **uncertainty** increase if I relax my **alpha** threshold?

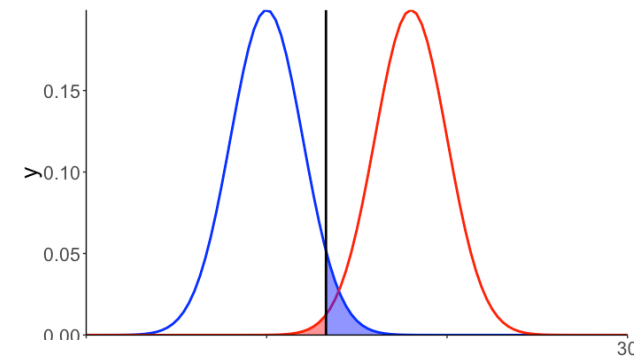
Notice how these are questions about **signal detection**, not practical meaningfulness!



alpha threshold



effect size



sample size

What α should we use? It's become conventional to set $\alpha = 0.05$. To be honest, this owes more to the fact that the resulting k tends to 1.96 as $n \rightarrow \infty$, and $1.96 \approx 2$, and most psychologists and economists could multiply by 2, even in 1950, than to any genuine principle of statistics or scientific method. A 5% error rate corresponds to messing up about one working day in every month, which you might well find high. On the other hand, there is nothing which stops you from increasing α . It's often illuminating to plot a series of confidence sets, at different values of α .

What about power? The **coverage** of a confidence set is the probability that it includes the true parameter value. This is not, however, the only virtue we want in a confidence set; if it was, we could just say "Every possible parameter is in the set", and have 100% coverage no matter what. We would also like the *wrong* values of the parameter to have a high probability of *not* being in the set. Just as the coverage is controlled by the size / false-alarm probability / type-I error rate α of the hypothesis test, the probability of excluding the wrong parameters is controlled by the power / miss probability / type-II error rate. Test with higher power exclude (correctly) more parameter values, and give smaller confidence sets.

How do we “turn the knobs”?

Power analysis via **simulation**

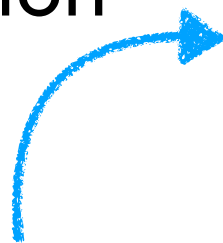
- Remember “Statistics for Hackers”?

- **Monte-carlo simulation**

- Bootstrapping

- Permuting

- Cross-validation



Recipe

- pick: α , n , *effect-size*
- simulate a large number of data sets of size n with the specified *effect-size*
- for each data set, run a statistical test to calculate the *p-value*
- calculate **sensitivity/power**: proportion of times we *correctly rejected* H_0 (given that H_1 is true)

Tomorrow & next week

Parameter Inference

Dataset: Credit-Card Debt

index	income	limit	rating	cards	age	education	gender	student	married	ethnicity	balance
1	14.89	3606	283	2	34	11	Male	No	Yes	Caucasian	333
2	106.03	6645	483	3	82	15	Female	Yes	Yes	Asian	903
3	104.59	7075	514	4	71	11	Male	No	No	Asian	580
4	148.92	9504	681	3	36	11	Female	No	No	Asian	964
5	55.88	4897	357	2	68	16	Male	No	Yes	Caucasian	331
6	80.18	8047	569	4	77	10	Male	No	No	Caucasian	1151
7	21.00	3388	259	2	37	12	Female	No	No	African American	203
8	71.41	7114	512	2	87	9	Male	No	No	Asian	872
9	15.12	3300	266	5	66	13	Female	No	No	Caucasian	279
10	71.06	6819	491	3	41	19	Female	Yes	Yes	African American	1350

variable	description
income	in thousand dollars
limit	credit limit
rating	credit rating
cards	number of credit cards
age	in years
education	years of education
gender	male or female
student	student or not
married	married or not
ethnicity	African American, Asian, Caucasian
balance	average credit card debt in dollars

Does **student** predict **credit card balance**?

Parameter Inference

```
1 # Create model
2 m_aug = model("Balance ~ Income", credit)
3
4 # Fit model (estimate parameters)
5 m_aug.fit()
6 m_aug.params
```

term str	estimate f64
Intercept	246.5147505914033
Income	6.04836340853156

```
1 # Perform parameter inference
2 m_aug.fit().infer().summary()
```

Linear Model: Balance ~ Income

Residuals:

Min	1Q	Median	3Q	Max
-803.642	-348.994	-54.425	331.748	1100.246

Coefficients:

	Estimate	Std.Err	t value	Pr(> t)
Intercept	246.515	33.199	7.43	6.90e-13 ***
Income	6.048	0.579	10.44	<2.2e-16 ***

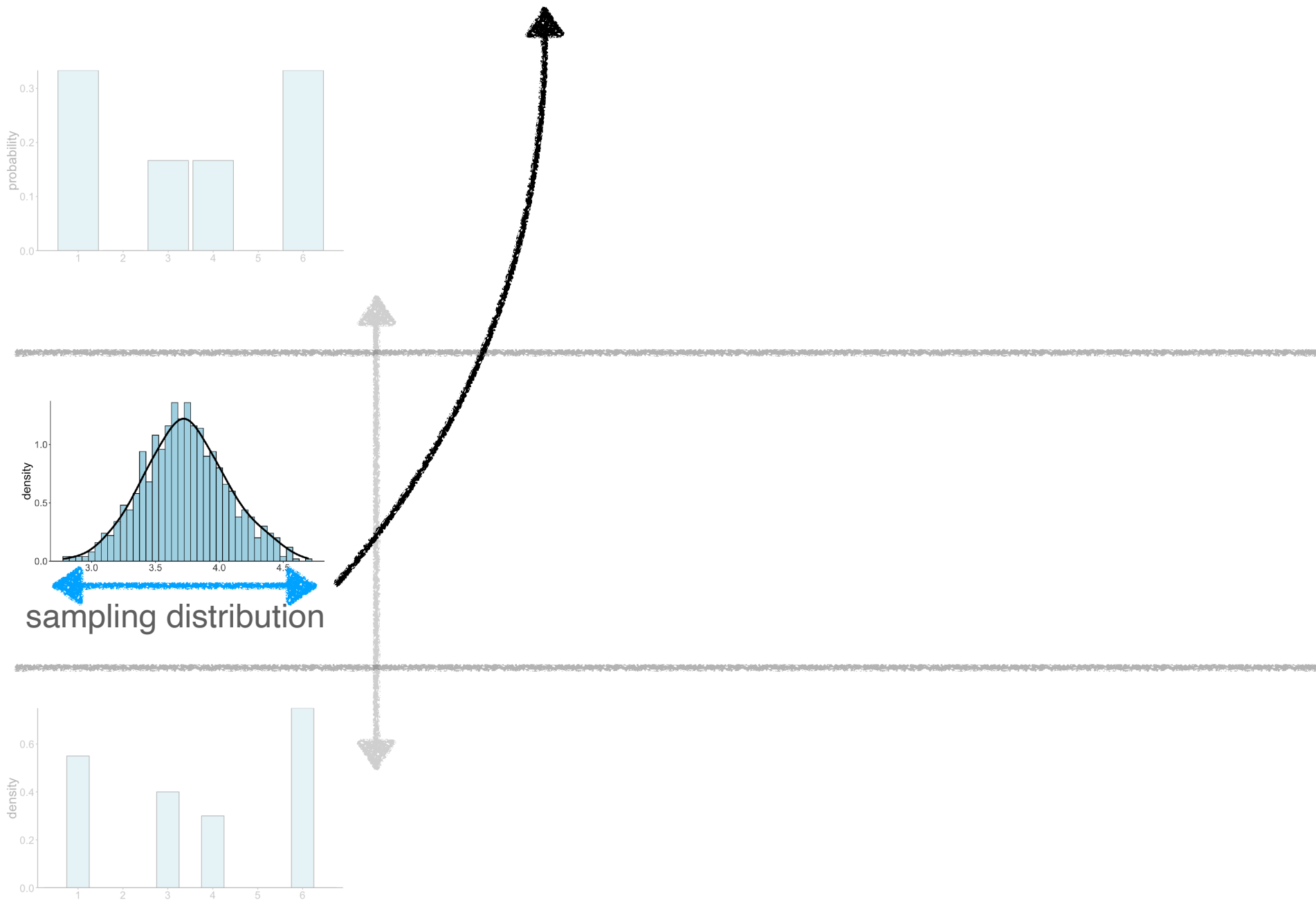
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual std error: 407.865 on 398 df

Multiple R-squared: 0.215, Adjusted R-squared: 0.213

AIC: 5945.9, BIC: 5953.9

Standard Error: uncertainty of estimate



$$\hat{Balance}_i = \beta_0 + \beta_1 * Income_i + \epsilon$$

Balance	Income
i64	f64
333	14.891
903	106.025
580	104.593
964	148.924
331	55.882
1151	80.18
203	20.996
872	71.408
279	15.125
1350	71.061

$$\begin{bmatrix} 333 \\ 903 \\ 580 \end{bmatrix} = \begin{bmatrix} 1 & 14.891 \\ 1 & 106.025 \\ 1 & 104.593 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}$$

y X β

2 unknown parameters
we're *estimating*

-
-
-

$$\hat{Balance}_i = \beta_0 + \beta_1 * Income_i + \epsilon$$

$$\hat{\beta} = \underbrace{(X^T X)^{-1}}_{\text{Similarity between predictors}} X^T y$$

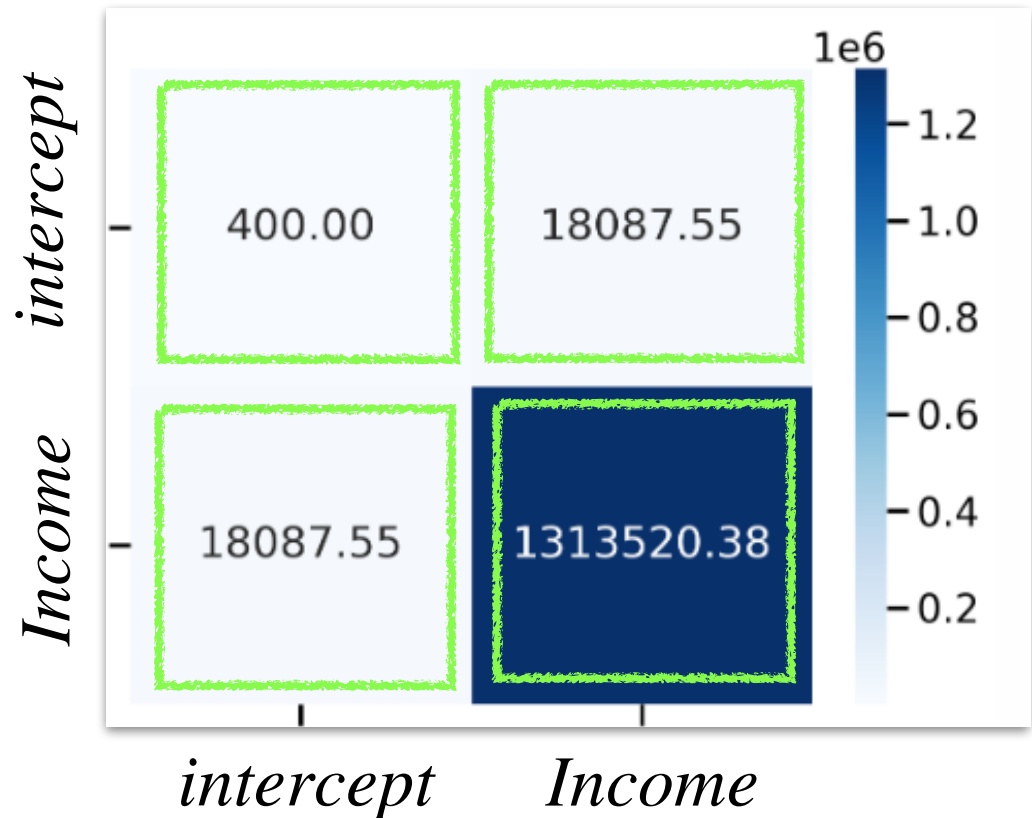
intercept *Income*

$$\begin{bmatrix} 1 & 14.891 \\ 1 & 106.025 \\ 1 & 104.593 \end{bmatrix}$$

X

Unscaled covariance
between predictors

Similarity between predictors
Unscaled variance of
each predictor



$$\hat{Balance}_i = \beta_0 + \beta_1 * Income_i + \epsilon$$

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

Remove **variance** of each predictor

$$SE_{\hat{\beta}} = \sqrt{\sigma^2 (X^T X)^{-1}}$$

$$= \frac{\sum residuals^2}{df}$$

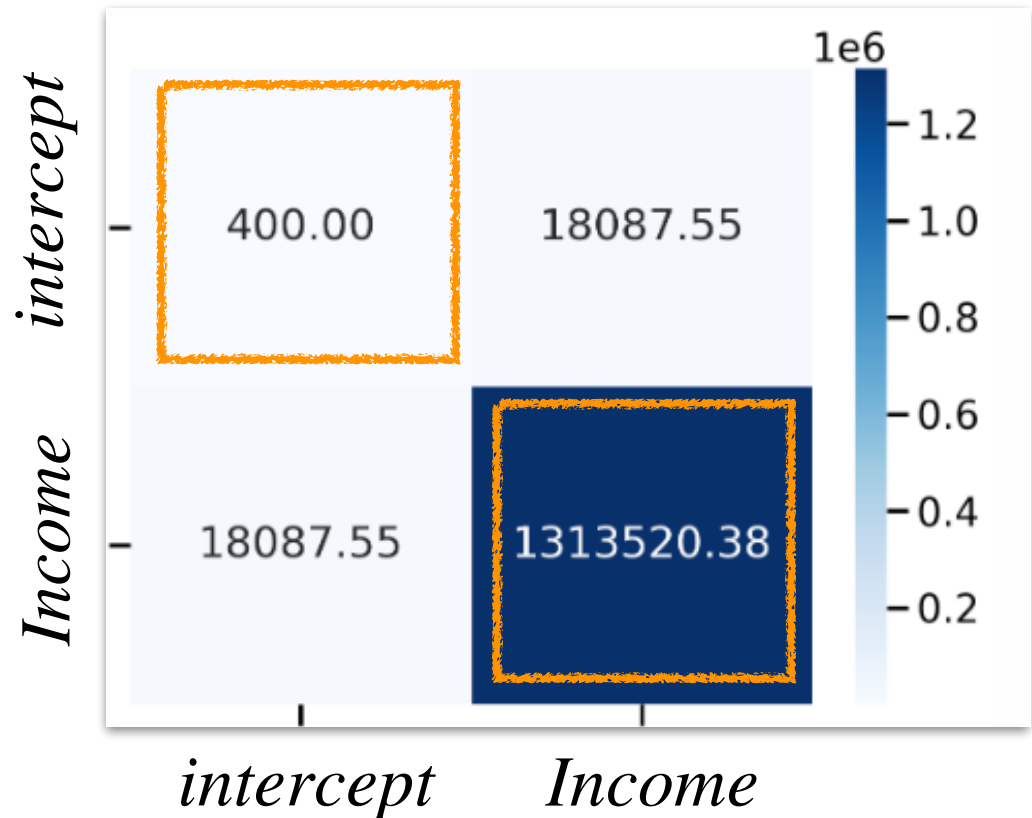
$$= \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{N - p}$$

Scale by

Variance of **residuals**

aka

Average **error** of the model



$$\hat{Balance}_i = \beta_0 + \beta_1 * Income_i + \epsilon$$

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

Remove **variance** of each predictor

$$SE_{\hat{\beta}} = \sqrt{\sigma^2 (X^T X)^{-1}}$$

$$= \frac{\sum residuals^2}{df}$$

$$= \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{N - p}$$

Scale by
Variance of **residuals**
aka

Average **error** of the model

```
np.sqrt(
    augmented_results.mse_resid * np.diag(np.linalg.inv(X.T @ X))
)
✓ 0.0s
array([33.19934735, 0.57935016])
```

$$\hat{Balance}_i = \beta_0 + \beta_1 * Income_i + \epsilon$$

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

$$SE_{\hat{\beta}} = \sqrt{\sigma^2 (X^T X)^{-1}}$$

```
np.sqrt(
    augmented_results.mse_resid * np.diag(np.linalg.inv(X.T @ X))
)
✓ 0.0s
array([33.19934735, 0.57935016])
```

Linear Model: Balance ~ Income

Residuals:

Min	1Q	Median	3Q	Max
-803.642	-348.994	-54.425	331.748	1100.246

Coefficients:

	Estimate	Std.Err	t value	Pr(> t)
Intercept	246.515	33.199	7.43	6.90e-13 ***
Income	6.048	0.579	10.44	<2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual std error: 407.865 on 398 df

Multiple R-squared: 0.215, Adjusted R-squared: 0.213

AIC: 5945.9, BIC: 5953.9

$$\hat{Balance}_i = \beta_0 + \beta_1 * Income_i + \epsilon$$

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

$$SE_{\hat{\beta}} = \sqrt{\sigma^2 (X^T X)^{-1}}$$

$$t = \frac{\hat{\beta}}{SE_{\hat{\beta}}}$$

$$t_{intercept} = \frac{246.51}{33.19} = 7.425$$

Linear Model: Balance ~ Income

Residuals:

Min	1Q	Median	3Q	Max
-803.642	-348.994	-54.425	331.748	1100.246

Coefficients:

	Estimate	Std. Err	t value	Pr(> t)
Intercept	246.515	33.199	7.43	6.90e-13 ***
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Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

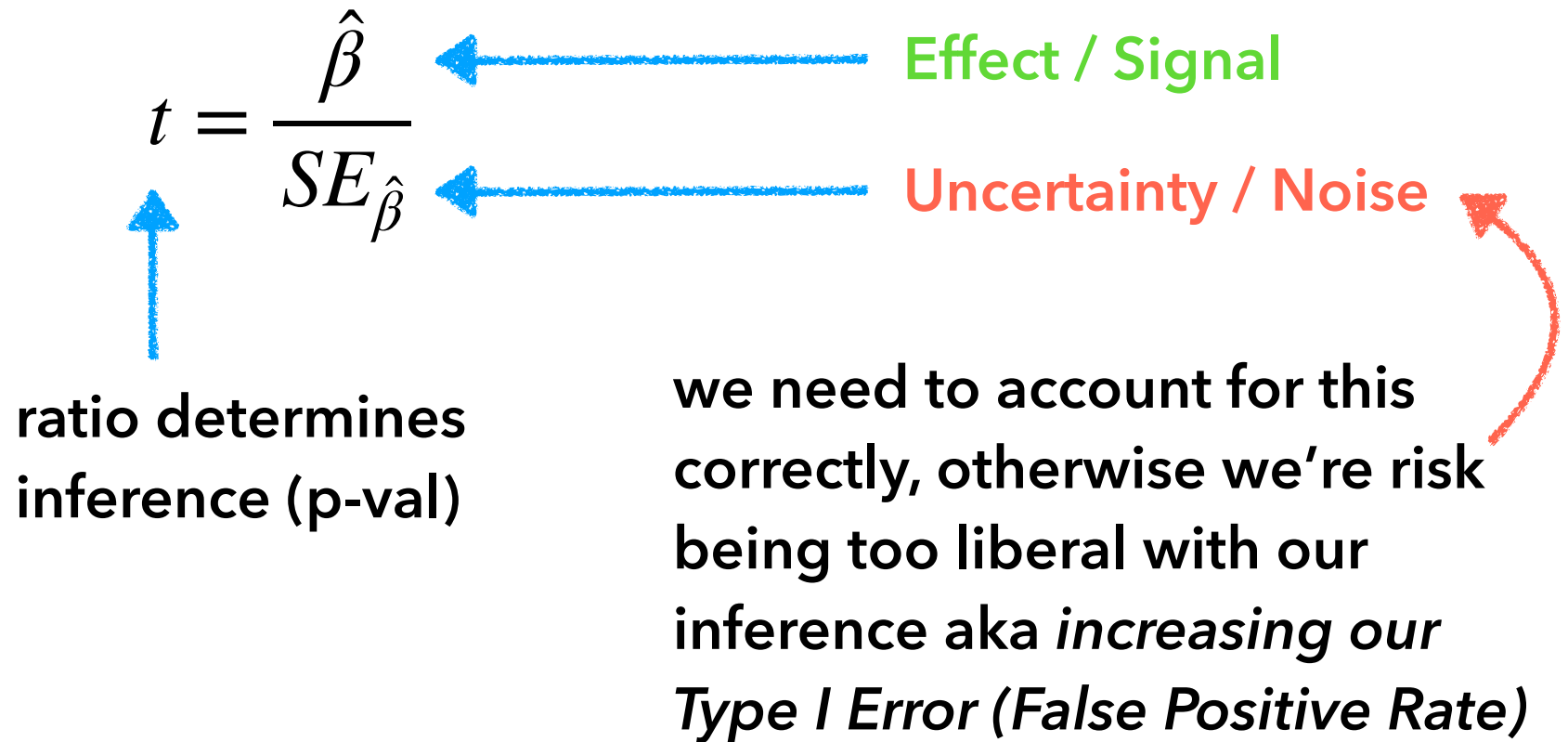
Residual std error: 407.865 on 398 df

Multiple R-squared: 0.215, Adjusted R-squared: 0.213

AIC: 5945.9, BIC: 5953.9

$$t_{Income} = \frac{6.04}{0.579} = 10.440$$

Parameter Inference



Parameter Inference

Linear Model: Balance ~ Income

Residuals:

Min	1Q	Median	3Q	Max
-803.642	-348.994	-54.425	331.748	1100.246

Coefficients:

	Estimate	Std.Err	t value	Pr(> t)
Intercept	246.515	33.199	7.43	6.90e-13 ***
Income	6.048	0.579	10.44	<2.2e-16 ***

```
1 # Create & fit compact (null) model
2 m_comp = model("Balance ~ 1", credit).fit()
3
4 # Compare
5 results = compare(m_comp, m_aug)
6 results
```

$$F = t^2$$

$$t = \sqrt{F}$$

model str	PRE f64	F f64	rss f64	ss f64	df f64	df_resid f64	p_value f64
Balance ~ 1			84,339,911.91			399	
Balance ~ Income	0.215	108.9917	66,208,744.5108	18,131,167.3992	1	398	0

Parameter Inference is proportional to nested model comparison!

$$F = t^2$$

$$t = \sqrt{F}$$

```
1 # Parameter t-statistic Squared
2 m_aug.params[1, 'statistic'] ** 2
```

✓ 0.0s

108.99171519122353

```
1 # Model comparison F-ratio
2 results[1, 'F']
```

✓ 0.0s

108.9917

Parameter Inference Takeaways

$$t = \frac{\hat{\beta}}{SE_{\hat{\beta}}}$$

- Large t-stat or small p-value is **not** about what variables are most important for prediction!
- It's about what **signals** can be discerned from **noise**
 - *Does the addition of this parameter provide lower prediction error than what adding random noise would produce?*
- Why? Because our *uncertainty* (SE) is influenced by
 - Magnitude of $(\hat{\beta})$
 - Model avg error (MSE)
 - Effective sample size (# observations vs # params aka DF)
 - Variance of each predictor ($x_1 \dots x_n$)
 - Correlation between predictors (columns of X)

Remember: Statistical significance is not practical significance

7.5.4 Appropriate Uses of p -Values and Significance Testing

I do not want this section to give the impression that p -values, hypothesis testing, and statistical significance are unimportant or necessarily misguided. They're often used badly, but that's true of every statistical tool from the sample mean on down the line. There are certainly situations where we really do want to know whether we have good evidence against some *exact* statistical hypothesis, and that's just the job these tools do. What are some of these situations?

Model checking Our statistical models often make very strong, claims about the probability distribution of the data, with little wiggle room. The simple linear regression model, for instance, claims that the regression function is *exactly* linear, and that the noise around this line has *exactly* constant variance. If we test these claims and find very small p -values, then we have evidence that there's a detectable, systematic departure from the model assumptions, and we should re-formulate the model.

Actual scientific interest Some scientific theories make very precise predictions about coefficients. According to Newton, the gravitational force between two masses is inversely proportional to the *square* of the distance between them, $\propto r^{-2}$. The prediction is exactly $\propto r^{-2}$, not $\propto r^{-1.99}$ nor $\propto r^{-2.05}$. Measuring that exponent and finding even tiny departures from 2 would be big news, if we had reason to think they were real and not just noise¹². One of the most successful theories in physics, quantum electrodynamics, makes predictions about some properties of hydrogen atoms with a theoretical precision of one part in a trillion; finding even tiny discrepancies between what the theory predicts and what we estimate would force us to rethink lots of physics¹³. Experiments to detect new particles, like the Higgs boson, essentially boil down to hypothesis testing, looking for deviations from theoretical predictions which should be exactly zero if the particle doesn't exist.

Outside of the natural sciences, however, it is harder to find examples of interesting, exact null hypothesis which are, so to speak, "live options". The best I can come up with are theories of economic growth and business cycles which predict that the share of national income going to labor (as opposed to capital) should be constant over time. Otherwise, in the social sciences, there's usually little theoretical reason to think that certain regression coefficients should be *exactly* zero, or *exactly* one, or anything else.

Turn the limit around: at least for large n ,

$$P_n \approx e^{-n \frac{\beta_1^2}{2\sigma^2/s_X^2}} \quad (7.57)$$

Thus, *any* $\beta_1 \neq 0$ will (eventually) give exponentially small p -values. This is why, as a saying among statisticians have it, “the p -value is a measure of sample size”: any non-zero coefficient will become arbitrarily statistically significant with enough data. This is just another way of saying that with enough data, we can (and will) detect even arbitrarily small coefficients, which is what we *want*. The flip-side, however, is that it’s just senseless to say that one coefficient is important because it has a really small p -value and another is unimportant because it’s got a big p -value. As we can see from Eq. 7.57, the p -value runs together the magnitude of the coefficient ($|\beta_1|$), the sample size (n), the noise around the regression line (σ^2), and how spread out the data is along the x axis (s_X^2), the last of these because they control how precisely we can estimate β_1 . Saying “this coefficient must be really important, because we can measure it really precisely” is not smart.

**Tomorrow we'll get back to
notebooks to set you up for HW 3!**